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Thermal hydraulic model of the molten salt reactor experiment with the NEAMS system analysis module ☆

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ABSTRACT

System analysis codes have a long history of providing best-estimate and conservative safety analysis for both light water and advanced reactor technologies, including molten salt reactors. As interest continues to expand with advanced reactor concepts, system analysis codes will need revisions to accommodate the behavior of these technologies. Legacy system analysis codes will need to be updated to the latest numerical techniques to shorten execution time and increase the accuracy of results. One example of a modern system analysis code that already encompasses these characteristics is the System Analysis Module (SAM).

One key objective of this paper was to review available information for system code modeling of the Molten Salt Reactor Experiment (MSRE) from sources in the open literature and collect the information from these open sources in one place for the first time. This supports the potential objective of developing an open specification for system code analysis for MSRE steady state and transients with and without reactor kinetics. Data from actual MSRE tests will serve as the basis for code-to-code comparison exercises, including the MSRE zero power physics tests, the fuel pump start-up and coast down tests, and the natural circulation transient. The objective is to produce a code-to-code benchmark with a standardized set of comparison problems, recognizing the limitations of the original data.

To demonstrate an initial application of this objective and the usefulness of compiling this open data, two Molten Salt Reactor Experiment (MSRE)-related models were developed to evaluate SAM for liquid fueled molten salt reactors. One model was the SAM MSRE hydraulic mockup, which provided experimental data for pressure drop measurements. The second model was the complete MSRE primary loop. The MSRE primary loop model incorporated a fluoride salt fuel/coolant with heat transfer in both the core and heat exchanger. For both the hydraulic mockup and MSRE primary loop models, a holistic 1-D system description was built using open documentation, an open description that can be readily modified and applied for any system analysis code.

SAM results for the pressure drop of the hydraulic mockup model were within 6% with measurements. Coolant temperatures for the primary loop model matched the expected axial change in temperature from historical calculations. Using alternative coolant properties obtained from the literature, corresponding to salts with different actinide contents, returned similar trends in core temperature profiles. A thermal hydraulic demonstration of a loss-of-flow transient showed the importance of coupling SAM thermal hydraulic analysis to neutronics. This coupling is essential for simulating MSR transients with system analysis codes.

1. Introduction

The concept of molten salt reactors (MSRs) has existed for over half a century. From 1965 to 1969 Oak Ridge National Laboratory (ORNL) maintained and operated the Molten Salt Reactor Experiment (MSRE). This program was a continuation of the Aircraft Reactor Experiment, which sought to study the use of molten fluoride salts for a circulatory fuel system that would power military aircraft (Rosenthal et al., 1970).

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The MSRE was composed of a primary loop in which the nuclear reaction took place and heated up the working fluid, which was coupled to a secondary loop via a shell-and-tube heat exchanger. The secondary loop served to remove heat produced from the primary loop to the environment via a radiator. The original fuel salt of the MSRE primary loop was LiF-BeF_2-ZrF_4 mixed with UF_4, while the secondary loop did not contain zirconium nor any actinides (Carbajo et al., 2017). Table 3 in Section 3.3 provides molecular compositions of three fuel salts used in the MSRE. The molten salt acts as both the fuel and coolant on the primary loop as it flows at 0.076 m^3/s through a network of Hastelloy-N pipes and 1140 parallel and independent fuel channels that are surrounded by graphite (Robertson, 1965). The fuel channels and graphite are exclusive to the core and are shown in Fig. 1. Fig. 2 shows an above view of a set of fuel channels surrounded by the graphite moderator. The figure shows the fuel channels as white ellipse-like holes surrounded by the crosshatched graphite stringer. While in the core, the coolant is heated from approximately 908 K to 936 K with 94% of the energy deposited into the fluid and 6% into the graphite (Carbajo et al., 2017). These values are from historical calculations (Haubenreich et al., 1964). The MSRE was designed to operate at 10 MW_t but actually operated at 8 MW_t.

Before and in parallel to operation of the facility, ORNL operated and maintained a full-scale hydraulic mockup of the MSRE. This facility was an open loop hydraulic mockup with no heat transfer or molten salt that was used for hydraulic testing of the system design. The hydraulic mockup maintained the dimensions and geometry of the MSRE. Both water and surrogate fluids were used in the hydraulic mockup. The surrogate fluids were also used because these liquids enhanced similitude relative to the fuel salt (Kedl, 1970). Fig. 3 below shows the hydraulic mockup core vessel.

MSRs have both challenges and potential benefits. Two significant potential benefits are near atmospheric operating pressure and coolant temperatures substantially below their boiling points. Additionally, fluoride salts have a high solubility for uranium and

Fig. 1. MSRE Graphite Core Being Assembled (Rosenthal, 2009).

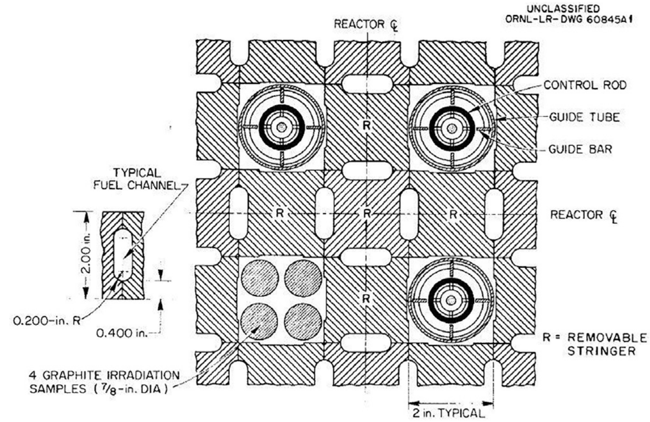

Fig. 2. MSRE Fuel Channel Arrangement Surrounded by Graphite and Control Rods (Haubenreich et al., 1964).

Fig. 3. MSRE Hydraulic mockup Core Vessel (Kedl, 1970).

exhibit inert behavior with common structural metals, including Hastelloy-N (Rosenthal et al., 1970). The fact that molten salt fuel is always in the liquid phase during normal operations also presents some advantages such as continuous online fission product removal and no fuel assembly fabrication or handling (He, 2016). Molten salt fuel in the MSRE also has a negative temperature coefficient of reactivity meaning that as the fuel/coolant temperature rises, power decreases (Rosenthal et al., 1970). These advantages are what promote the safety qualities of MSRs.

Challenges with MSRs were highlighted in a U.S. Department of Energy (DOE) sponsored study, the Advanced Demonstration and Test Reactor (ADTR) Options Study. This study compared different advanced reactor technologies and assessed them against four high level strategic objectives that would characterize advanced reactor technology options (Petti et al., 2017). The challenges for MSRs regard licensability, the chemical properties of actinide- and fission product- bearing salts (Petti et al., 2017), coolant corrosion, and overall lesser maturity of this rector type compared to reactors such as Sodium-cooled Fast Reactors (SFRs) and High Temperature Gas-cooled Reactors (HTGRs).

Most system analysis codes, such as RELAP5-3D or TRACE, were developed targeting light water reactors (LWRs). New simulation tools are currently under development that are oriented towards modeling advanced reactors that contain single-phase coolants. One such tool is the U.S. DOE Nuclear Energy Advanced Modeling and Simulation (NEAMS) tool SAM. The objective of SAM development was to create a code that is user-friendly and fast while improving system transient analysis for advanced reactor technologies. This means that SAM can solve coupled reactor phenomena, such as fission and heat transfer, with current numerical capabilities at quicker speeds than legacy programs (Hu et al., 2015). It also utilizes the Multi-physics Object Oriented Simulation Environment (MOOSE) (Hu, 2017). More specifically, SAM has been applied to simulate and assess Sodium-cooled Fast Reactors, Lead-cooled Fast Reactors, Fluoride-salt-cooled High temperature Reactors, and Molten Salt Reactors. All of these advanced reactor designs utilize single-phase, low-pressure, and high-temperature coolants, which are advantageous from a reactor safety perspective (Hu et al., 2015). SAM has been designed for these reactor types with built-in equation of state (EOS) modeling for the different coolants and component types that accurately depict the functionality of advanced reactors. The code also utilizes high-order finite element modeling for single-phase fluid flow and heat transfer with flexible coupling between fluid and solid components of the reactor system (Hu, 2017). SAM developments are underway for a reduced-order multi-dimensional flow model and the code is currently capable of 3-D flexible core modeling and computational fluid dynamics (CFD) integration.

Interest in MSRs has led to its study with various system analysis codes. One such study models the MSRE with TRACE for dynamic simulations of the experiment (He, 2016). This model provided accurate results when compared to experimental data for steady state and transient behavior (e.g. it demonstrated a step insertion of reactivity, a ramp insertion of reactivity, and a primary pump failure) (He, 2016). Krepel et al. (2007) simulated the effective loss of delayed neutrons in steady state operation, fuel pump startup, fuel pump coast-down, and natural circulation using the DYNamical 3-Dimensional code for MSRs (DYN3D-MSR). This spatial code showed that the pump startup and coastdown agreed well with experimental data from the MSRE, but differences existed with tracking delayed neutrons and natural circulation of the primary system which could be resolved by including an initial reactivity surge early in the transient (Krepel et al., 2007).

Additionally, RELAP5 has been used in multiple studies to model the MSRE. Shi et al. (2016) analyzed the point kinetics model with the effects of flowing fuel with an internal heat source in the RELAP/SCDAPSIM/MOD4.0 code, a modified version of RELAP5. One result from this work showed a loss of reactivity as delayed neutron precursors were carried outside of the core and into the primary loop (Shi et al., 2016). Additionally, Carbajo et al. (2017) used RELAP5-3D to depict both the MSRE hydraulic mockup and primary loop. This work showed reasonable agreement between code and experiments for coolant velocity and pressure drop of the hydraulic mockup. It also showed similar agreement between code and historical calculations for coolant temperatures (among other parameters) of the primary loop MSRE.

1.1. Objectives of the present study

The study of MSRE thermal hydraulic behavior with SAM is the central objective of this paper. The MSRE program produced publicly available thermal hydraulic data that was either measured or calculated. This project takes this data and uses it for SAM MSRE models. This legacy data is not ideal for the validation of a modern computational tool. However, there are no open specifications for reactor safety problems related to MSRs. The first objective of this paper is to produce a detailed description of the MSRE primary system based on public data that can be used as a basis to develop an open specification for code-to-code comparison of MSR safety analysis tools. These specifications would focus on actual tests performed in MSRE, for example the fuel pump start-up and coast down tests, among others. A major contribution of this paper is a 1-D primary loop system description of MSRE that can be adopted for any system analysis tool.

This paper is divided into two main parts. Section 2 discusses the SAM model description for the hydraulic mockup and results obtained for system pressure drop. This also includes a discussion on other parameters, such as coolant velocity. Section 3 focuses on the SAM model description for the MSRE primary loop and results obtained for the axial core coolant temperature profile. Other scenarios are incorporated with SAM to study varying coolant properties and a transient demonstration of a loss-of-flow (LOF) accident.

All of the results obtained from SAM are compared to measurements or calculations given in original MSRE documentation. This paper demonstrates that the data generated during the time the MSRE program was active can be used as a basis of historical comparison with a modern system analysis code for thermal hydraulic analysis.

2. Hydraulic mockup model

2.1. Model description

The SAM hydraulic mockup model is constructed of pipes connected together by junctions or branches. The SAM pipe component simulates 1-D fluid flow in a cylindrical pipe and heat conduction in the pipe wall (Hu et al., 2015). SAM branch or junction components connect pipes together. Branch components, which are zero dimension junctions where multiple 1-D components can be connected, are used when loss coefficients must be specified. A junction is used when loss coefficients are unnecessary. Fig. 4 shows a general nodalization of the SAM hydraulic mockup model as demonstrated in this work. The diagram is not to scale, and the coordinate location of the inlet of each component is listed. SAM requires that each component's inlet position be specified. The origin of the model is set at the inlet of the component titled "core1."

The initial coolant velocity of the hydraulic mockup model is set to 6.0 m/s to reflect a flow rate of 0.076 m^3/s (Robertson, 1965). Initial conditions for pressure (0.101 MPa) and temperature (300 K) are also assumed in the SAM calculation. The density of water is assumed to be 997 kg/m^3 with a dynamic viscosity of 8.51e-4 kg/m-s. SAM also requires input of specific heat capacity, internal enthalpy, and thermal conductivity, but these values are irrelevant since the working fluid is not heated in this case.

All pipe components in the SAM MSRE hydraulic mockup are Schedule 40 type with a wall thickness of 0.0066 m. These are all composed of carbon steel with a cross-sectional area of 0.0127 m^2 with hydraulic diameter of 0.127 m (Robertson, 1965). The lengths of all pipes are unique and are described in Robertson (1965). The pipe roughness for all pipes in the hydraulic mockup is assumed to be 1.00e-4 m. The SAM branch and junction components that connect the pipes have unique values for flow area and inlet/outlet loss coefficients to account for changes in cross-sectional area between pipes. The values for the flow areas are assumed to be consistent with that of the downstream pipe.

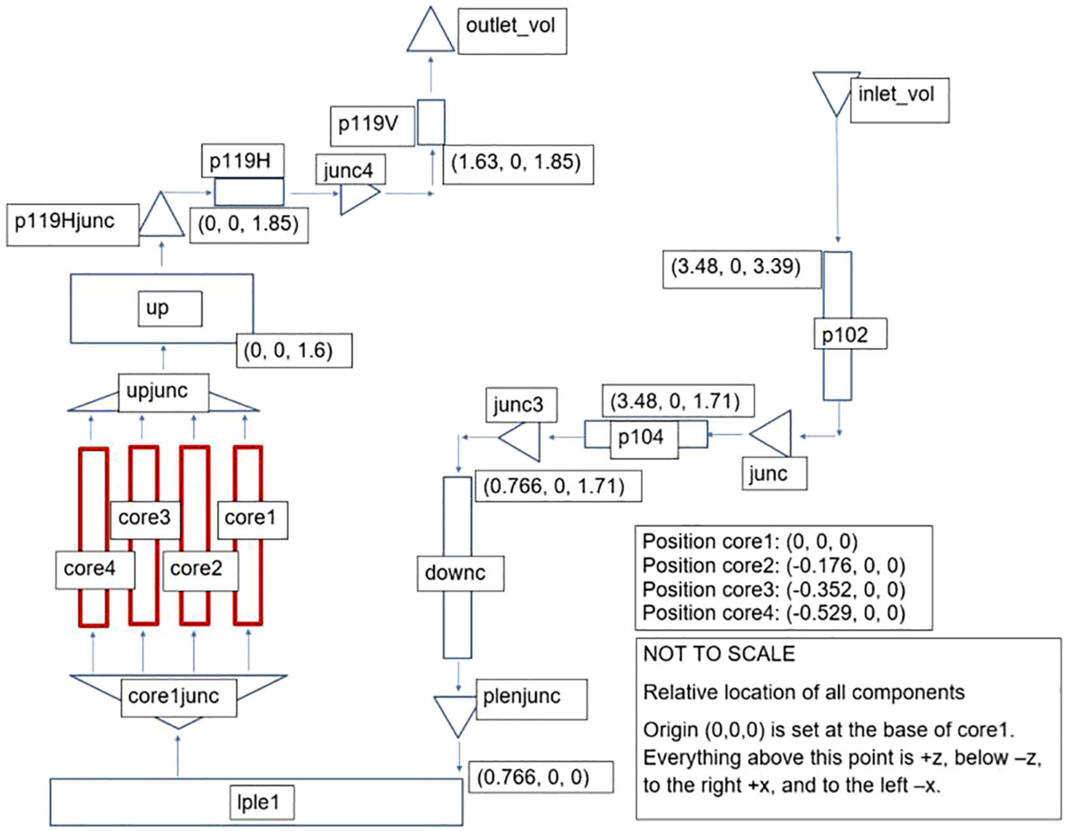

Fig. 4. MSRE Hydraulic Mockup Model Nodalization.

The values for the inlet and outlet loss coefficients are calculated based on geometry using formulas from Todreas and Kazimi (2012). Table 1 shows the loss coefficients that have been calculated for each of the components in the hydraulic mockup.

All components in the SAM MSRE hydraulic mockup model are depicted with 1-D flow. It is important to note that the MSRE downcomer is a 3-D annulus that propels the working fluid into a lower plenum so that it can flow upwards through the core channels. The downcomer flow contains radial, angular, and axial velocity components (Carbajo et al., 2017), making it a complex component. A complex flow emanates from the volute, or flow distributor, located atop the core vessel containing several 3/4'' diameter holes at 30° angles with the vertical (Robertson, 1965). This is seen in Fig. 5. The downcomer is modeled as a 1-D pipe on SAM, which is expected to lead to some inaccuracies. The annular geometry of the downcomer allows for a hydraulic diameter of 0.0508 m. The cross-sectional area is 0.116 m^2 based on Haubenreich et al. (1964).

The lower plenum is complex. This component is unique in that its height has been minimized to 0.01 m to account for complex directional shifting flows observed here. The lower plenum transitions the flow from vertical downward to vertical upward over a short length. It is not clear how much vertical displacement the fluid will have in the lower plenum since it could be minimal (neglecting a lower plenum) or maximal (including the entire

Table 1
MSRE Hydraulic Mockup Loss Coefficients.

Component	Inlet Loss Coefficient	Outlet Loss Coefficient
Inlet Horizontal Piping	Not Used	0.397
Downcomer	0.496	0.869
Lower Plenum	0	0.649
Core Rings	16.7	17.0
Upper Plenum	0.447	0.985
Outlet Horizontal Piping	0.0	Not Used

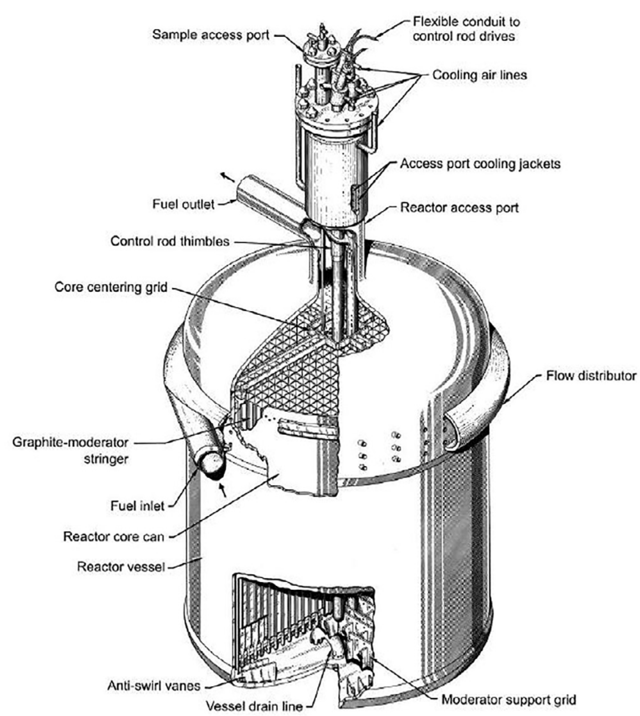

Fig. 5. MSRE Core Vessel Showing Volute and Distribution Holes (Rosenthal, 2009).

length of the lower plenum). The hydraulic diameter and flow area of the lower plenum are 1.47 m and 1.71 m^2, respectively (Haubenreich et al., 1964). Additionally, the upper plenum has identical hydraulic diameter and flow area as the lower plenum, but its height differs, 0.249 m per Haubenreich et al. (1964).

The final contribution to the modeling description is a discussion on the MSRE fuel channels. Although they are not carrying fuel in the hydraulic mockup, channels are still present in the aluminum core structure of this facility. The total cross sectional area of all 1140 channels is 0.332 m^2 with hydraulic diameter of 0.0159 m (Carbajo et al., 2017). All channels have a height of 1.60 m per Haubenreich et al. (1964). These values were confirmed using dimensions for individual channels reported in Haubenreich et al. (1964). The channels are divided into four regions for power distribution purposes that will be observed in the MSRE primary loop model. These four regions, or rings, are input as pipe components in SAM. Two hydraulic mockup models are constructed with different assumptions for the core ring flow areas. One model has varying flow areas consistent with the RELAP5-3D model built by Carbajo et al. (2017), while the other has constant flow areas for all rings. The position of each core ring is based on the inner radius of the MSRE core vessel, which is given as 0.705 m (Haubenreich et al., 1964). Each core ring's center is laterally spaced by 0.176 m, the thickness of each ring (the total radius divided by four). This process simplifies radial geometry of the MSRE to 1-D geometry necessary for SAM modeling.

2.2. SAM hydraulic mockup analysis

Kedl (1970) provides pressure drop measurements for the hydraulic mockup. These measurements include the total system pressure drop. This data from Kedl (1970) provides a basis of historical comparison for the SAM hydraulic mockup model. This comparison is conducted using each of the two SAM hydraulic mockup models discussed in the previous section. Additionally, the SAM calculations are repeated over a range of flow rates.

Reasonable alignment shown in Fig. 6 between code and experiment indicate that the geometry has been modeled correctly using SAM. This plot shows the total system pressure drop over a range of flow rates for two different geometries. Measurement uncertainties were not reported by Kedl (1970). Fig. 7 shows a plot comparing the relative percent difference between the SAM models and experimental data. The two SAM models are expected to behave nearly identically since loss coefficients are changed in accordance with the changes to flow area. All other flow conditions and geometries outside of the core remain the same for both SAM models.

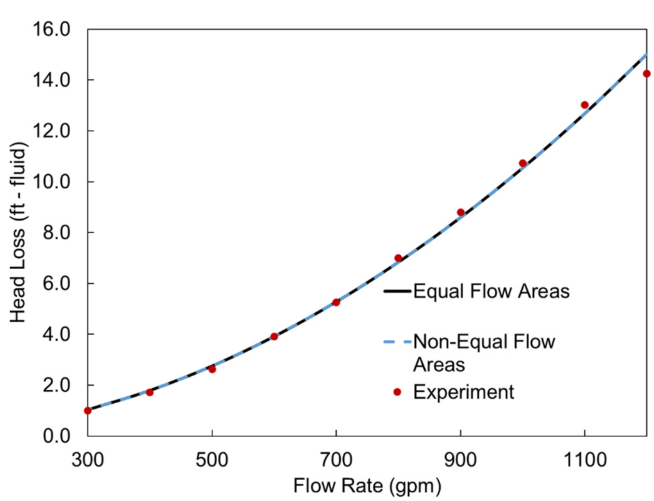

Fig. 6. Hydraulic Mockup Head Loss Comparison with Measured Values.

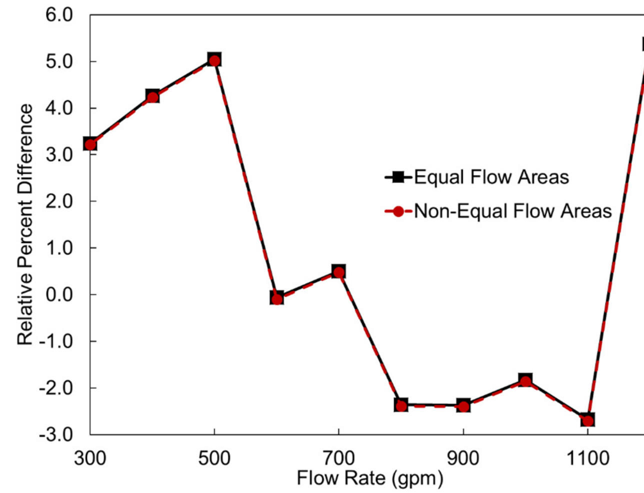

Fig. 7. Percent Difference Comparison with Measured Values.

Kedl (1970) shared other hydraulic parameters, such as the coolant velocity. Most velocities align well between the code and experiment in all regions of the hydraulic mockup with exception of the downcomer. SAM predicts a coolant velocity at the bottom of the downcomer of 0.66 m/s whereas Kedl (1970) reported a value of 1.68 m/s. While the value from SAM is consistent with a RELAP5-3D model of the hydraulic mockup (Carbajo et al., 2017), it significantly differs from the measured value. The difference can be explained by looking at the geometry of the downcomer. As mentioned in Section 2.1, the MSRE hydraulic mockup has complex 3-D flow in this area while SAM only models this phenomenon in 1-D. CFD modeling is necessary to capture this behavior. Nonetheless, reasonable agreement between the SAM model and measurements for pressure drop and a majority of velocities gives a reasonably representative basis for comparison of the SAM hydraulic mockup for geometry input over a range of coolant flow rates.

3. MSRE primary loop model

3.1. Model description

The SAM model of the MSRE primary loop contains the same pipe network geometry and dimensions as the hydraulic mockup. One difference between the facilities is material composition of components. The pipes in MSRE were composed of Hastelloy-N rather than carbon steel to prevent structural corrosion with the molten salt fuel/coolant. The core structure was composed of graphite rather than aluminum. The primary loop also contains additional components and features that are not present in the hydraulic mockup. This model has a cover gas modeled as a SAM pressure outlet with pressure 0.1 MPa, the approximate system pressure of the MSRE, to protect the fuel/coolant from contact with air or moisture. The core is now heated from approximately 908 K to 936 K by making use of the SAM volumetric heat source function. Decay heat is assumed to contribute to the heating in the core region and is not distributed along the remaining primary loop. A heat exchanger is added with a simplified secondary loop that serves as a heat sink for the heated fuel/coolant. The SAM built-in properties option for FLiBe is activated as a close estimate to the actual fuel/coolant of the MSRE. Sensitivity analysis on built-in FLiBe properties versus those actually used in MSRE are presented in Section 3.3. A fuel pump with discharge head of

0.092 MPa is used to circulate the salt around the entire primary loop at a rate of 171 kg/s, per Robertson (1965).

Heated components are essential in the SAM MSRE primary loop model. During MSRE operation, the core power was divided between the fuel/coolant and graphite structure, with 94% of the thermal power deposited into the fuel/coolant and the remainder in the graphite (Carbajo et al., 2017). For the SAM model, it is assumed that all power (10 MW_t) is deposited into the fuel/coolant. Even if only 94% of the power were deposited into the fuel/coolant, 100% of the heat generated would need to be removed to give the same change in temperature. Since a volumetric heat source is required by SAM, the power deposited into the fuel/coolant is divided by the fluid volume of the four rings giving 1.88e7 W/m^3. A typical cosine power profile is assumed in the four core rings using 20 axial nodes. The volumetric heat source that reflects the power profile is input equally for each of the four core rings. The reason to use four core rings is to verify that the code is providing consistent results for each ring. These rings could also be used later on to implement a radial power distribution in the core.

The heat exchanger used in the MSRE had a U-tube design with a primary side shell and secondary side tubes, both made of Hastelloy-N. The heat exchanger is horizontally oriented at 1.83 m in length (Kedl and McGlothlan, 1968). Table 2 displays values obtained from Kedl and McGlothlan (1968) or calculated from other data. Calculations use equations from Lee (2010).

The surface area density, listed in Table 2, is a key parameter needed when simulating heat transfer with SAM. This parameter is the ratio of the heated surface area to fluid volume. Additionally, the heat transfer coefficients, not listed in Table 2, are adjusted to achieve the expected change in temperature for both the primary and secondary loop coolants. The expected change in temperature is 28 K and 41 K for the primary and secondary coolants, respectively (Engel and Haubenreich, 1962). This preliminary approach gives satisfactory results for the primary loop without considering the complete geometry details of the heat exchanger or assessing the appropriateness of different heat transfer correlations. The SAM model also includes a simplified and preliminary representation of the secondary side of the heat exchanger. The use of slightly different fluid properties between SAM and MSRE also contributes to this preliminary approach.

The flow rate on the secondary loop is 0.054 m^3/s, per Robertson (1965), which sets the inlet coolant velocity to 1.3 m/s. The secondary loop inlet coolant temperature is set to 825 K while outlet pressure is set to 0.1 MPa. These are boundary conditions that remain unchanged. This simplified secondary loop of the MSRE serves only as a heat sink to complete the power balance of the system.

Table 2
MSRE Heat Exchanger Dimensions.

Parameter	Value	Reference
Triangular Pitch (m)	0.0197	Kedl and McGlothlan (1968)
Outer Tube Diameter (m)	0.0127	Kedl and McGlothlan (1968)
Tube Thickness (m)	0.00107	Kedl and McGlothlan (1968)
Inner Tube Diameter (m)	0.0106	Calculated
Tube Flow Area (m^2)	0.0279	Calculated
Baffle Spacing (m)	0.305	Kedl and McGlothlan (1968)
Outer Shell Diameter (m)	0.406	Kedl and McGlothlan (1968)
Shell Thickness (m)	0.0127	Kedl and McGlothlan (1968)
Inner Shell Diameter (m)	0.381	Calculated
Tube Clearance (m)	0.00699	Calculated
Shell Flow Area (m^2)	0.0412	Calculated
Tube Hydraulic Diameter (m)	0.0209	Calculated
Shell Hydraulic Diameter (m)	0.0130	Calculated
Tube Surface Area Density (m^{-1})	378	Calculated
Shell Surface Area Density (m^{-1})	308	Calculated

3.2. Primary loop model analysis

The SAM MSRE model features steady state analysis using the system described in Section 3.1. The primary results discussed in this section are the fuel/coolant temperatures in the core and heat exchanger (including secondary loop coolant of the heat exchanger). Due to limited instrumentation on the MSRE, calculations were necessary to estimate the expected coolant temperatures. According to Haubenreich et al. (1964), the specific calculation method involved the flow distribution of the hydraulic mockup combined with the power-density distribution prediction. Any referenced values for the MSRE primary loop are historical calculations and not measurements.

The axial coolant temperature profile for the core is shown in Fig. 8. The core coolant temperature profile is identical for all core rings due to equal geometry and power, and qualitatively behaves as expected. However, there is slight variation with the expected results for the core inlet and outlet. The SAM model shows a core coolant temperature of 905 K at the inlet and 933 K at the outlet. These are slightly different than the historical calculation of 908 K at the inlet and 936 K at the outlet (Engel and Haubenreich, 1962). Although both the temperatures differ by 3 K from the calculation, the relative temperature change is consistent with the literature (28 K). The FLiBe-based salt properties used in SAM differ from those of the actual MSRE coolant and may contribute to the slight variation in temperature. A more likely scenario is that the heat exchanger causes the slight temperature difference. Since heat is removed with this component, slightly lower coolant temperatures are caused by some of the approximations assumed for the heat exchanger.

Fig. 9 shows expected behavior for both sides of the heat exchanger. For the primary side, the axial coolant temperature is expected to decrease from 936 K to 908 K. For the secondary side, the axial coolant temperature is expected to increase from 825 K to 866 K (Engel and Haubenreich, 1962). SAM results show that the primary loop working fluid is cooled from 933 K to 905 K, as expected, with a 3 K offset. For the secondary side, the coolant temperature increases from 825 K to 882 K. The outlet temperature differs from the expected value by 16 K. The model of the secondary side of the heat exchanger is highly simplified in this work. These calculations for the secondary loop are deemed acceptable, as the focus of this work is on the modeling of the primary loop.

Other parameters of interest when analyzing the MSRE primary loop model include system pressure and coolant velocity. The system pressure is shown next to the plot for the core axial pressure

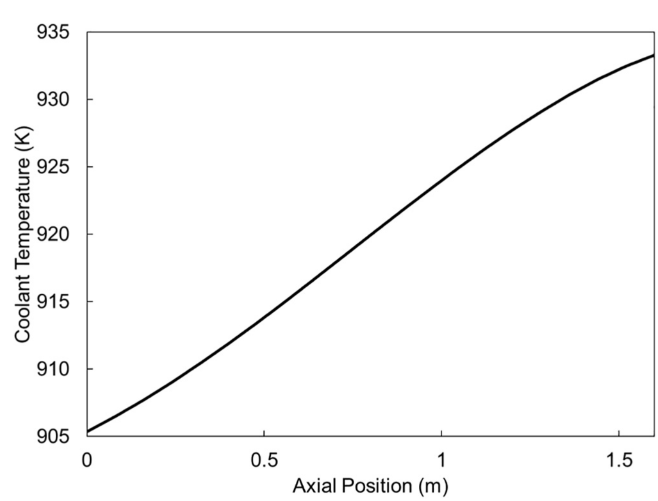

Fig. 8. SAM MSRE Primary Loop Model Axial Core Coolant Temperature Profile.

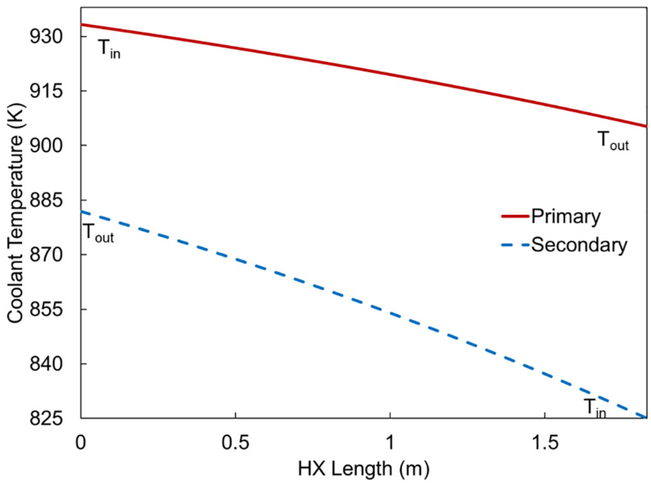

Fig. 9. SAM MSRE Primary Loop Model Heat Exchanger Temperature Profiles.

drop in Fig. 10. The system pressure is depicted using a ParaView (Ayachit, 2017) visualization of the pressure change throughout the components of the MSRE model. The plot provides a more quantitative assessment by showing the expected linear change in pressure throughout the length of one of the core rings. Furthermore, Engel and Haubenreich (1962) gives the expected core coolant velocity as a range from 0.18 m/s to 0.61 m/s to account for the different concentric rings. The results from SAM give a core velocity at both the inlet and outlet of 0.23 m/s for all rings since they are of equal flow area. Although this value falls within the range given by Engel and Haubenreich (1962), it is well short of the average of this range, 0.40 m/s. This difference is likely caused by the different flow areas used between the SAM model and the original experiment for the core rings. The SAM model utilizes equal flow areas while the original MSRE experiment had non-equal flow areas for the core rings.

In addition to comparing with legacy calculations, the results from Carbajo et al. (2017) are compared with the results obtained with SAM. Table 3 shows this comparison between the two codes and historical results for both the hydraulic mockup and MSRE primary loop. The historical results for the MSRE primary loop (calculations) are from Engel and Haubenreich (1962) and those for the hydraulic mockup (measurements) are from Kedl (1970). Reasonable agreement implies that the MSRE 1-D primary loop model description can be modified and applied to other system analysis tools.

Table 3
Comparison of SAM MSRE Primary Loop and Hydraulic Mockup Parameters with Other Works.

	SAM	RELAP5-3D	Historical Calculations
Core Coolant T_{in} (K)	905	908	908
Core Coolant T_{out} (K)	933	936	936
Coolant Flow Rate (m^3/s)	0.076	0.077	0.076
Core Velocity (m/s)	0.23	0.20–0.50	0.18–0.61
Inlet Pipe Velocity (m/s)	5.87	5.85	5.85
Downcomer Velocity (m/s)	0.66	0.66	1.68
Core Head Loss (kPa)	1.91	1.78	1.79
System Head Loss (kPa)	44.9	44.7	44.8

3.3. Sensitivity to the salt properties

All of the previous results presented in this paper utilize SAM built-in properties for FLiBe. Although using the SAM built-in properties for FLiBe gives reasonable results, Robertson (1965) provided properties for three different salts used during original MSRE testing. These salts all had different concentrations of constituent molecules, as shown in Table 4. The three salts are labeled in this table as "MSRE 1", "MSRE 2", and "MSRE 3". Robertson (1965) specified that "MSRE 1" was a thorium-uranium blend of fuel. It goes on to state that "MSRE 2" was composed of highly enriched uranium and "MSRE 3" was composed of partially enriched uranium. Additionally, the thermophysical properties of these salts are provided as constants evaluated at 922 K. These three salts are simulated in the MSRE primary loop model and compared to SAM built-in FLiBe EOS.

Applying the properties for the different salts in SAM created various results for the axial core coolant temperature profile. These profiles are shown in Fig. 11. The figure shows that the temperature profiles all exhibit similar trends, but there are noticeable quantitative differences. The MSRE salts all have higher coolant temperatures when compared to SAM built-in FLiBe properties. Moreover, the MSRE salts all exhibit a higher coolant temperature gradient than SAM built-in FLiBe properties, roughly 32 K rather than the reference's 28 K. The differences in the temperature profiles are mainly attributed to the change in actinide concentrations and ZrF_4 content between SAM built-in EOS for FLiBe and MSRE salts. SAM built-in EOS for FLiBe does not contain actinides nor ZrF_4. This results in different hydraulic and thermal properties. Table 5 shows that the density and viscosity are higher and the specific heat capacity is lower for MSRE salts when comparing to SAM built-in FLiBe EOS. The equations that describe the SAM built-in FLiBe EOS are found in Hu (2017).

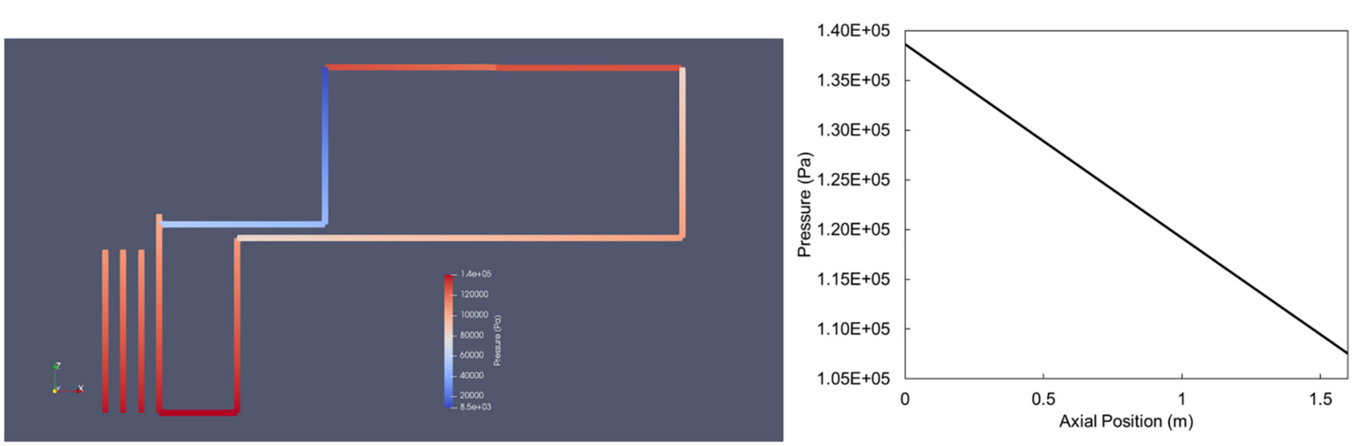

Fig. 10. ParaView Pressure Visualization (Left) with Core Ring Axial Pressure Drop (Right).

Table 4
Molecular Composition of Different Fuel Salts.

	Molecular Composition (%)				
	LiF	BeF_2	ZrF_4	ThF_4	UF_4
MSRE 1	70.0	23.6	5.0	1.0	0.4
MSRE 2	66.8	29.0	4.0	0.0	0.2
MSRE 3	65.0	29.1	5.0	0.0	0.9

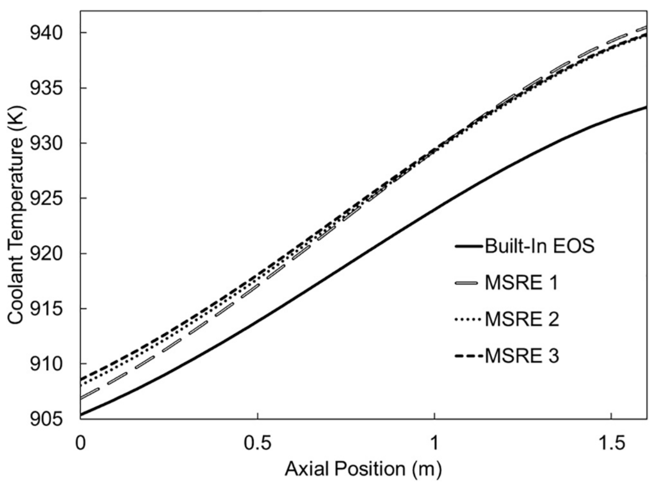

Fig. 11. Axial Core Coolant Temperature Profile Comparison of Different Fuel Salts.

Table 5
Comparison of Thermophysical Properties between SAM Built-in FLiBe EOS and MSRE Salts.

	Density (kg/m^3)	Viscosity (kg/m-s)	Specific Heat Capacity (J/kg-K)
Built-in FLiBe	1963	0.00661	2416
MSRE 1	2243	0.00744	1883
MSRE 2	2082	0.00703	2008
MSRE 3	2146	0.00827	1966

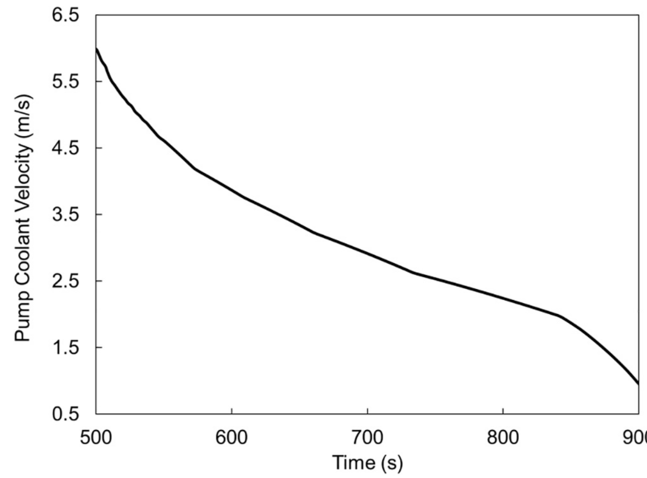

Fig. 12. Coolant Velocity of Fuel Pump following Start of LOF.

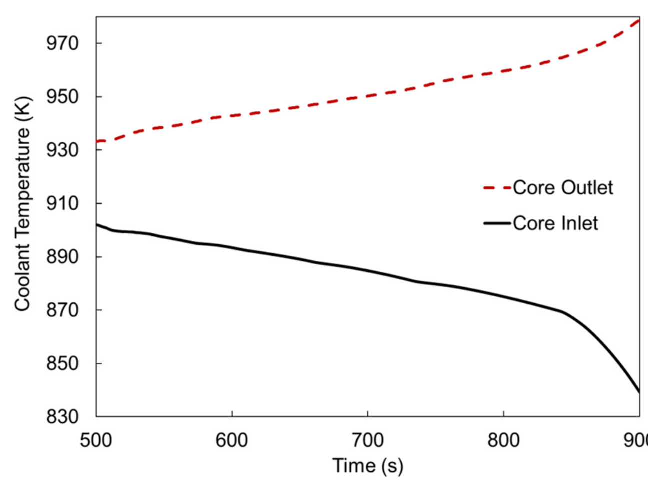

Fig. 13. Axial Core Coolant Temperatures following Start of LOF.

3.4. Transient demonstration

A key step in safety analysis simulation is to understand the reactor and system response to an accident scenario. For MSRs, one of the possible design basis accidents (DBA) is a flow perturbation that leads to a loss-of-flow (LOF). A pump coastdown curve is applied to the fuel pump component in the SAM model to represent a pump failure, a possible cause of a LOF. The pump curve used in this work is derived from Gao et al. (2010). SAM uses this pump curve as a time dependent function that is triggered at 500 s, after steady state conditions have been reached. In the SAM model, the pipe components that represent the core rings are substituted with channel components to model the graphite stringer that surrounds the fuel channels. Energy deposition is now divided between the coolant and graphite. This is an acceptable modeling approach since channel components are needed to simulate any graphite related reactivity feedback that might occur during a transient or accident. It is important to note that this simulation neglects reactivity feedback and impact from a SCRAM event.

Fig. 12 shows the results for the fuel pump coolant velocity following the flow perturbation. The plot of the fuel pump coolant velocity confirms that the pump trip has been implemented correctly. This transient response does not reach a natural circulation state. Plotting the coolant velocity indicates that more information could be obtained if this work were to be coupled with neutronics feedback. This infers that delayed neutron precursors can be monitored as they travel around the entire MSRE primary system.

Fig. 13 shows the core coolant inlet and outlet temperature response to the pump coastdown. Once the transient begins, the core inlet temperature decreases while the core outlet temperature increases. The secondary side of the heat exchanger shows a slight increase in outlet temperature (882 K during steady state to 887 K during transient response). The change in temperature of the secondary side coolant has changed by 5 K, a modest difference that results in a 9.2% increase in the heat exchanger's heat removal rate. An increase in the heat removal rate would lead to a decrease in core inlet coolant temperature. The secondary side of the heat exchanger has a fixed coolant inlet temperature of 825 K since the entire secondary loop is not modeled. In reality, the temperature of the secondary coolant would start increasing and would not reset to an initial value, which is what is currently assumed in this model. This simplification is not what occurs physically but explains the large change in temperature observed in the primary loop. Additionally, flow stagnation is observed in the four core rings and primary side of the heat exchanger. The exact reason for this has not been investigated in this work. Further, when the fuel pump starts to fail, the mass flow rate of the primary loop coolant decreases implying that the change in coolant temperature must increase to conserve power. It is also important to note that the convective heat transfer coefficients have been approximated without considering the geometry details of the heat exchanger. This approximation is reasonable for steady state but will need to be reassessed for full transient analysis. This demonstration shows that SAM can model a postulated accident.

4. Summary and future work

The modern system analysis code SAM is targeting advanced reactor designs, such as SFRs and MSRs. SAM utilizes modern flexible numerical methods, has second order accuracy, and has several options to assist in resolving numerical instabilities. This work provides a basis for using SAM to model the MSRE. This work also contributes two detailed MSRE model descriptions, one for the hydraulic mockup and one for the molten salt based primary loop.

Measured data exists for the open flow hydraulic mockup, and approximate temperatures were calculated during original MSRE operation for the FLiBe-based MSRE primary loop. Steady state data from both the hydraulic mockup and primary loop are compared to results obtained with SAM to perform a preliminary comparison. Additional studies are also conducted to further demonstrate that the model developed with SAM behaves in a physically accurate manner under various conditions. This work demonstrates the applicability of a recent NEAMS tool for comparative analysis with legacy MSRE data.

The major achievements of this work can be broken into three main categories:

1. A holistic 1-D MSRE system description of both the hydraulic mockup and molten salt based primary loop that can be used for any system analysis code.
2. A SAM MSRE hydraulic mockup model with results for total system head loss. Comparisons are made between SAM results and measured data to compare the code with hydraulic mockup historical data.
3. A SAM MSRE primary loop model with results for core and heat exchanger coolant axial temperature profiles. Comparisons are made between SAM results and original MSRE calculations to compare the code with the FLiBe-based primary loop. Additional studies show that various fluid properties and a transient can be demonstrated with reasonable accuracy.

This work also leaves open the possibility for several areas of recommended future work. CFD, whether coupled or independent from SAM, could be used to model the complex 3-D flows observed in the MSRE downcomer and develop a reduced order model of this feature. Additional complexity can be added to the SAM model to simulate radial power or varying number of core rings. Further investigations into the bias seen with the pressure drop in the hydraulic mockup and the salt properties in the MSRE primary loop are necessary to select the most accurate initial conditions of the SAM simulation. Additionally, the transient demonstration of a LOF shows that neutronics coupling is necessary. Neutronics and thermal hydraulic coupling can assist to better model delayed neutron precursor drift and reactivity feedback changes in MSRs due to changes in temperature. This capability is now available in SAM, but was not demonstrated in this work (Zhang and Hu, 2018). Overall, this coupling will further safety analysis of MSRs with system analysis codes.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.anucene.2018.10.060.

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